# Spillover-Pandemic Pipeline: Detailed Mathematical Specification

## Notation

| Symbol | Meaning |
|---|---|
| $c$ | Country index |
| $i$ | Spillover event index |
| $t$ | Year |
| $j, \ell$ | Cell indices within a 2D swath |
| $g$ | Generation index in the branching process |
| $\Lambda_c(t)$ | Number of spillover events in country $c$ in year $t$ (from ecological model) |
| $R_0$ | Basic reproduction number for a given spillover virus |
| $k$ | Dispersion parameter of the negative binomial offspring distribution |
| $N_j$ | Total population of cell $j$ |
| $S_j(g)$ | Susceptible population in cell $j$ at generation $g$ |
| $I_j(g)$ | Number of actively infectious individuals in cell $j$ at generation $g$ |
| $T_g$ | Generation time (~3 days for influenza) |
| $d_{j\ell}$ | Distance between cells $j$ and $\ell$ (km) |
| $\beta$ | Distance decay rate in gravity kernel (per km) |
| $\alpha$ | Population attractiveness exponent in gravity kernel |
| $w_{\text{self}}$ | Self-transmission weight (inflates within-cell transmission) |
| $N_{\text{crit}}$ | Cumulative infections in hub cell required for PanDORA handoff |
| $\mathcal{H}$ | Set of hub cells within the swath |

---

## Stage 0: Offline Pre-Processing (done once per country)

### 0a. Spillover location probability surface

For each country $c$, we construct a probability surface $\pi_c(j)$ over grid cells $j$ that gives the relative probability

that a spillover event occurs in cell $j$. This is proportional to livestock density:

$$\pi_c(j) = \frac{L_c(j)}{\sum_{j' \in c} L_c(j')}$$

where $L_c(j)$ is the livestock density (animals per km²) in cell $j$, from the GLW3 raster. For influenza, the relevant livestock are primarily poultry (chickens, ducks) for avian influenza pathways, or pigs for swine influenza reassortment. The choice of species should reflect the spillover pathway being modelled. If multiple species are relevant, a weighted combination can be used:

$$L_c(j) = \sum_s w_s \cdot L_c^{(s)}(j)$$

where $L_c^{(s)}(j)$ is the density of species $s$ and $w_s$ are weights reflecting the relative contribution of each species to spillover risk. These weights could be informed by the relative frequency of different influenza subtypes in the ecological model's output, or set to reflect expert judgement. Note that $\pi_c(j) = 0$ for cells outside the country or on water.

**0b. Pre-compute 2D swaths**

For each country, sample $M$ spillover locations from $\pi_c$. For each sampled location $j_0$:

1. Identify the nearest urban hub $h$ (from a database of cities with population $\geq$ some threshold, or airports with international connectivity).
2. Compute the axis vector from $j_0$ to $h$ and extract all grid cells within a rectangular swath of half-width $W$ (e.g., 50 km) centred on this axis, extending from slightly behind $j_0$ to slightly past $h$.
3. For each cell $j$ in the swath, record:
   - Coordinates (lat, lon)
   - Population $N_j$ (from population density raster $\times$ cell area)
   - Whether $j \in \mathcal{H}$ (hub cell set, defined as cells within some radius $r_h$ of the hub centre, e.g., $r_h = 15$ km)
4. Compute and store the pairwise distance matrix $d_{j\ell}$ for all cells in the swath.

Store these swaths. At simulation time, each spillover event draws a pre-computed swath from this library. If a spillover location falls very close to or within a hub ($d_{j_0,h} < r_h$), flag it as "at-hub" — these events skip the spatial branching process (Stage 3b) and proceed directly to a non-spatial branching process at the hub.

---

## Stage 1: Draw Viral Properties

For each spillover event $i$:

**1a. Draw $R_0$**

$$R_0^{(i)} \sim \text{LogNormal}(\mu, \sigma)$$

where $\mu$ and $\sigma$ are parameters of the log-normal distribution (i.e., the mean and standard deviation of $\log R_0$). The distribution should satisfy:

- **Median** $\ll 1$: most spillover viruses transmit poorly. A median of $\tilde{R}_0 \approx 0.1 - 0.3$ is consistent with observed zoonotic influenza behaviour. Since the median of a LogNormal is $e^{\mu}$, this implies $\mu \approx \ln(0.1)$ to $\ln(0.3) \approx -2.3$ to $-1.2$.
- **Right tail extends past 1**: the probability $P(R_0 > 1)$ should be small but non-negligible. Given historical pandemic frequency (~4 per 125 years) and global spillover rates, this probability will be very small (order $10^{-4}$ to $10^{-2}$ depending on global spillover volume). The parameter $\sigma$ controls the weight in this tail: larger $\sigma$ gives a heavier tail and more frequent $R_0 > 1$ draws.
- **Tail should reach 2–3**: the 1918 pandemic had $R_0 \approx 2 - 3$. The distribution should assign non-zero probability to this range.

**Computational shortcut:** If $R_0^{(i)} < R_{\text{min}}$ (e.g., $R_{\text{min}} = 0.05$), transmission is negligible. Record the event as a single human case and skip to mortality tallying. This avoids running a branching process for the vast majority of events.

**1b. Draw or fix the dispersion parameter $k$**

$$k^{(i)} = k_0 \quad (\text{fixed, e.g., } k_0 = 1)$$

or, if desired:

$$k^{(i)} \sim \text{some distribution}$$

The offspring distribution for each infected individual in the branching process is:

$$\text{offspring} \sim \text{NegBin}(\text{mean}, k)$$

where $\text{NegBin}(m, k)$ has probability mass function:

$$P(X = x) = \binom{x + k - 1}{x} \left(\frac{k}{k + m}\right)^k \left(\frac{m}{k + m}\right)^x$$

The variance is $m + m^2/k$. When $k \to \infty$, this approaches a Poisson with mean $m$. Smaller $k$ means more overdispersion (superspreading). For influenza, $k \approx 0.5 - 2$ is the commonly estimated range.

---

## Stage 2: Route Determination

Given $R_0^{(i)}$:

| Condition | Route |
|---|---|
| $R_0 < R_{\text{min}}$ (~0.05) | Single case. Skip to Stage 5. |
| $R_{\text{min}} \leq R_0 < 1$ | Non-spatial branching process (Stage 3a). Chain always dies out. |
| $R_0 \geq 1$ and spillover is at-hub | Non-spatial branching process at hub (Stage 3c). No spatial bottleneck, but stochastic extinction still possible. |
| $R_0 \geq 1$ and spillover is not at-hub | Spatially explicit branching process (Stage 3b). |

---

## Stage 3a: Non-Spatial Branching Process ($R_0 < 1$)

When $R_0 < 1$, the chain dies out with probability 1 regardless of spatial structure. We need only the total chain size $C$ to compute mortality.

### Simulation approach

Starting from $I(0) = 1$, at each generation $g$:

$$I(g+1) = \sum_{n=1}^{I(g)} X_n, \quad X_n \sim \text{NegBin}(R_0, k)$$

Continue until $I(g) = 0$. The total chain size is $C = \sum_g I(g)$.

### Analytical shortcut

The total chain size $C$ from a single ancestor in a branching process with offspring mean $R_0 < 1$ has a known distribution. The probability generating function of $C$ satisfies $G_C(s) = s \cdot G_X(G_C(s))$ where $G_X$ is the pgf of the offspring distribution. For the negative binomial:

$$G_X(s) = \left( \frac{k}{k + R_0(1-s)} \right)^k$$

The expected chain size is:

$$E[C] = \frac{1}{1 - R_0}$$

For the full distribution of $C$, one can use the Borel-Tanner distribution or simulate directly (which is fast since chains are short when $R_0 < 1$).

**Output:** total chain size $C^{(i)}$. Proceed to Stage 5.

---

### Stage 3b: Spatially Explicit Branching Process ($R_0 \geq 1$, not at hub)

This is the core of the pipeline. Load the pre-computed swath for this spillover event.

**Initialisation**

$$S_j(0) = N_j \quad \forall j$$

$$I_{j_0}(0) = 1, \quad I_j(0) = 0 \quad \forall j \neq j_0$$

$$C_j^{\text{hub}}(0) = 0 \quad \text{(cumulative infections in hub cells)}$$

where $j_0$ is the spillover cell.

**Generation $g \to g+1$**

For each cell $j$ with $I_j(g) > 0$, and for each of the $I_j(g)$ infected individuals:

**Step A — Draw number of offspring:**

$$n_{\text{offspring}} \sim \text{NegBin}\left(R_0 \cdot \frac{S_j(g)}{N_j},\ k\right)$$

The effective reproduction number in cell $j$ at generation $g$ is $R_0 \cdot S_j(g)/N_j$, which declines as the local cell is depleted. When $S_j(g) = 0$, the effective $R$ is zero and no further local transmission occurs.

**Step B — Assign each offspring to a destination cell via gravity kernel:**

For each of the $n_{\text{offspring}}$ offspring, independently draw a destination cell $\ell$ from the probability distribution:

$$P(\ell \mid j) = \frac{w_{j\ell}}{\sum_m w_{jm}}$$

where the unnormalised weights are:

$$w_{j\ell} = \begin{cases}
N_\ell^\alpha \cdot e^{-\beta \cdot d_{j\ell}} + w_{\text{self}} & \text{if } \ell = j \\[6pt]
N_\ell^\alpha \cdot e^{-\beta \cdot d_{j\ell}} & \text{if } \ell \neq j
\end{cases}$$

This is a gravity-type kernel with exponential distance decay rate $\beta$ and population attractiveness exponent $\alpha$. The $w_{\text{self}}$ term inflates within-cell transmission to reflect the dominance of household and close-contact transmission.

**Important:** the gravity weights can be pre-computed for each swath since they depend only on cell populations and distances, not on the dynamic state of the outbreak (the susceptible depletion enters through the effective $R_0$ in Step A, not through the kernel).

**Step C — Update state:**

For each offspring assigned to cell $\ell$:

- If $S_\ell(g) > 0$: the transmission is successful.
  - $I_\ell(g+1) \mathrel{+}= 1$
  - $S_\ell(g) \mathrel{-}= 1$
  - If $\ell \in \mathcal{H}$: $C^{\text{hub}} \mathrel{+}= 1$
- If $S_\ell(g) = 0$: the transmission fails (no susceptibles left in that cell). The offspring is lost.

After processing all infected individuals, set $I_j(g) = 0$ for the previous generation (they have recovered or been removed — this is a generation-based SIR, not continuous time).

**Termination conditions**

After each generation, check:

1. **Extinction:** $\sum_j I_j(g+1) = 0$. The outbreak has died out. Record total chain size $C = \sum_g \sum_j I_j(g)$. Proceed to Stage 5 with outcome "spatially contained."
2. **Hub establishment:** $C^{\text{hub}} \geq N_{\text{crit}}$. The outbreak has established in the urban hub. Proceed to Stage 4 (PanDORA handoff).
3. **Generation cap:** $g > g_{\max}$ (e.g., $g_{\max} = 100$). Terminate and classify as a large contained outbreak. This is a safety valve; in practice, outbreaks that haven't reached the hub by generation 50–100 are very unlikely to do so.

**Computational notes**

- **Pre-compute the kernel.** The gravity weights $w_{j\ell}$ depend only on the swath geometry and can be computed once and stored as a sparse matrix (since most weights are negligible for distant cells). At each generation, you only need to sample from the relevant row of this matrix.
- **Vectorise across individuals within a cell.** Rather than looping over each infected individual in cell $j$, draw the total number of offspring from cell $j$ as a sum of $I_j(g)$ independent NegBin draws. The sum of $n$ independent $\text{NegBin}(m, k)$ variables is $\text{NegBin}(n \cdot m, n \cdot k)$... actually, this is not quite right. The sum of $n$ iid $\text{NegBin}(m, k)$ is $\text{NegBin}(m, n \cdot k)$, which preserves the mean ($n \cdot m$) but changes the variance. Alternatively, draw the total offspring count as a sum, then multinomially assign them to destination cells using the kernel weights. Specifically:

$$N_{\text{total offspring from cell } j} = \sum_{n=1}^{I_j(g)} X_n, \quad X_n \sim \text{NegBin}\left(R_0 \cdot \frac{S_j(g)}{N_j}, k\right)$$

Then distribute these offspring across destination cells:

$$(\text{count to cell } \ell_1, \ldots, \text{count to cell } \ell_D) \sim \text{Multinomial}(N_{\text{total}},\ \{P(\ell \mid j)\}_\ell)$$

This avoids looping over individual offspring and is much faster.

---

### Stage 3c: Non-Spatial Branching Process at Hub ($R_0 \geq 1$, spillover at hub)

If the spillover occurs within or very near a hub cell, there is no spatial bottleneck. Run a standard (non-spatial) branching process from 1 infected individual with offspring distribution $\text{NegBin}(R_0, k)$. Since the hub population is very large, susceptible depletion is negligible in the early phase.

Track cumulative infections. If cumulative infections reach $N_{\text{crit}}$, hand off to PanDORA (Stage 4). Otherwise, if the chain goes extinct, record total chain size and proceed to Stage 5.

The probability of extinction from a single introduction, even with $R_0 > 1$, is the solution $q$ to:

$$q = G_X(q) = \left(\frac{k}{k + R_0(1-q)}\right)^k$$

For example, with $R_0 = 1.5$ and $k = 1$: $q \approx 0.67$ — about two-thirds of introductions go extinct even at the hub. With $R_0 = 2.0$ and $k = 1$: $q = 0.5$.

---

### Stage 4: PanDORA Handoff

When cumulative hub infections reach $N_{\text{crit}}$ (either from Stage 3b or 3c):

**Inputs to PanDORA:**

- Seeding country: $c$
- Initial infections: $N_{\text{crit}}$ (or total cumulative infections across all cells at the moment $N_{\text{crit}}$ is reached in the hub, whichever is preferred)
- $R_0$: the value drawn in Stage 1a

PanDORA simulates global spread and returns pandemic mortality $M^{(i)}_{\text{pandemic}}$.

---

### Stage 5: Mortality Tallying

For each spillover event $i$, the mortality is:

$$M^{(i)} = \begin{cases}
\text{IFR} & \text{if } R_0 < R_{\min} \text{ (single case)} \[4pt]
C^{(i)} \cdot \text{IFR} & \text{if chain died out (Stages 3a, 3b, or 3c)} \[4pt]
M_{\text{pandemic}}^{(i)} & \text{if handed off to PanDORA (Stage 4)}
\end{cases}$$

where $\mathrm{IFR}$ is the infection fatality rate. For sub-pandemic events, this could be:

- A fixed IFR (e.g., 0.01 for a typical influenza-like illness, or higher for H5N1-type viruses)
- A function of $R_0$ (if we believe that virulence and transmissibility are correlated)
- Drawn from a distribution alongside $R_0$

For pandemic events, PanDORA's internal IFR handles mortality calculation.

---

## Stage 6: Monte Carlo Aggregation

For a given year $t$:

### 6a. Draw spillover counts

For each country $c$, the ecological model provides $\Lambda_c(t)$ (the number of spillover events). If the ecological model provides a distribution rather than a point estimate, draw from it:

$$\Lambda_c(t) \sim \mathrm{Poisson}(\hat{\lambda}_c(t))$$

or use the point estimate directly.

### 6b. Simulate all events

For each of the $\sum_c \Lambda_c(t)$ events:

1. Sample a spillover location from $\pi_c$ and load a pre-computed swath.
2. Run Stages 1–5 to obtain mortality $M^{(i)}$.

### 6c. Aggregate

The total mortality in year $t$ for one Monte Carlo realisation is:

$$M_{\text{total}}(t) = \sum_c \sum_{i=1}^{\Lambda_c(t)} M^{(i)}$$

### 6d. Repeat

Repeat steps 6a–6c $K$ times (e.g., $K = 1000$ or more) to build a distribution of $M_{\text{total}}(t)$. Report:

- $E[M_{\text{total}}(t)]$ — annualised expected mortality
- Percentiles (5th, 50th, 95th) of $M_{\text{total}}(t)$
- $P(\text{at least one pandemic in year } t)$
- Conditional on pandemic: expected pandemic mortality

**6e. Across years**

Repeat for each year $t$ in the simulation horizon (e.g., 2000–2100). This produces time series of expected mortality, pandemic probability, etc., which can be analysed for trends driven by the ecological model's projected changes in spillover rates.

---

## Parameter Summary

### Parameters from data (fixed)

| Parameter | Source | Notes |
|---|---|---|
| $\Lambda_c(t)$ | Ecological model | Country-level spillover counts per year |
| $\pi_c(j)$ | GLW3 livestock rasters | Spillover location probability surface |
| $N_j$ | WorldPop / GPW | Population per cell |
| $d_{j\ell}$ | Cell coordinates | Pairwise distances within swath |
| Hub locations | GHSL / OurAirports | Major urban centres / international airports |

### Parameters from literature (informed priors)

| Parameter | Plausible range | Key references |
|---|---|---|
| $k$ | 0.5 – 2.0 | Influenza superspreading literature |
| $\beta$ | 0.05 – 0.15 per km | Gravity model literature; rural mobility data |
| $\alpha$ | 0.5 – 1.0 | Gravity model literature |
| IFR | 0.001 – 0.05 (sub-pandemic) | Varies by assumed virulence |

### Parameters to calibrate

| Parameter | Role | Calibration target |
|---|---|---|
| $\mu$ | Location of $R_0$ distribution (log-scale) | Historical pandemic frequency (~4 per 125 yr) |

| Parameter | Role | Calibration target |
|---|---|---|
| $\sigma$ | Shape/spread of $R_0$ distribution (log-scale) | Historical pandemic frequency + severity range |
| $w_{\text{self}}$ | Self-transmission weight | Can be jointly calibrated or fixed from contact data |
| $N_{\text{crit}}$ | Hub establishment threshold | Sensitivity analysis; results should be robust to 100–1000 range |

**Calibration constraint**

Given hindcast spillover rates $\{\Lambda_c(t)\}_{t=1900}^{2025}$ (from the ecological model, extrapolated backwards if needed), the expected number of pandemics over this period is:

$$E[\text{pandemics}] = \sum_{t=1900}^{2025} \sum_{c} \Lambda_c(t) \cdot P(\text{pandemic} \mid \text{spillover in } c)$$

where $P(\text{pandemic} \mid \text{spillover in } c)$ is the probability that a single spillover event in country $c$ passes through the entire pipeline to PanDORA. This probability depends on $\mu$, $\sigma$, $k$, $\beta$, $\alpha$, $w_{\text{self}}$, $N_{\text{crit}}$, and the spatial structure of country $c$. Setting $E[\text{pandemics}] \approx 4$ constrains the parameter space.